E-commerce Sales Trend & Customer Analysis

One-page business brief | InternCircle Project 03 | Synthetic 2025 order data

Context

This analysis reviews 2,000 simulated orders from 300 customers across four product categories in 2025. The dataset is generated for practice and is not real company or customer data.

Key findings

Measure	Result
Total simulated revenue	2,404,233.29
Average order value	1,202.12
Highest revenue month	2025-02 (232,027.20)
Lowest revenue month	2025-08 (179,783.75)
Top revenue category	Fashion (35.5% of revenue)
Top 30 customers (top 10%)	18.9% of total revenue

Recommendations

1. Launch a measured loyalty/VIP test for the top 10% of customers, who account for 18.9% of simulated revenue. Track repeat purchase rate and retention before and after the campaign.

2. Review stock availability and contribution margin for Fashion, the highest-revenue category (35.5% of simulated revenue), before increasing marketing investment.

3. Use the monthly revenue chart to plan campaigns ahead of the weakest month (2025-08) and investigate what contributed to the strongest month (2025-02). Measure campaign uplift against a baseline.

Next steps

Validate these patterns with authorised real business data before making operational decisions. Add profit margin, returns, acquisition channel, and customer cohort analysis in a future iteration.

Data note: Generated with NumPy random generator seed 42. Results are reproducible with the included notebook.